



Stockholm University

DIFO 2023 Lab-2 Answers and Forensic Report

Group-11

Group members

STUDENT_SUID
sabe4953
moha3312
khar1482

STUD_NAME
Samir Hossain Santo Bepu
Mohammad Jahid Hasan
Khondaker Refai Arafat

STUD_EMAIL
sabe4953@student.su.se
moha3312@student.su.se
khar1482@student.su.se

LAB 2 (Part-1)

Q01: What does su do? What is a significant thing that su allows you to do that you can't do otherwise?

Answer: The 'su' command stands for "Superuser" or "Switch User" in Unix-like operating systems, including Android. It is a command-line utility that allows a user to change their user identity and temporarily become another user or superuser (usually the root user) without logging out of their current session, enabling them to perform administrative tasks or run commands with different user privileges. It can also be used to run a specific command as another user.

Here are a few significant things we can do with su that we typically can't do otherwise:

- a. Installing and updating or removing system packages and software.
- b. Managing system configurations files, such as those in /etc, requires root access.
- c. Managing user accounts and passwords: Creating, deleting, or modifying user accounts and their password settings tasks.
- d. Network configuration: Configuring network interfaces, firewall rules, and other network-related tasks often demand root access.
- e. Managing system services: Starting, stopping, and configuring system services (e.g., web servers, databases) generally require superuser privileges.
- f. File and directory permissions: Changing permissions and ownership of system files and directories necessitates root access.

Q02: What is /dev used for in Linux?

Answer: In Linux, the /dev directory is a crucial part of the file system, and it stands for "device." It contains special files that represent and interact with physical and virtual devices on the system. These files are used for various device-related operations, including input/output (I/O) operations and device communication

Device Files: The /dev directory contains special device files, also known as device nodes or simply nodes. These files are used to access and communicate with hardware devices, such as hard drives, USB drives, printers, keyboards, and more.

Permissions: Device files have specific permissions that determine who can read from or write to them. Access to certain devices may be restricted to users with appropriate permissions.

For example, when you list the contents of /dev, you may see entries like /dev/sda representing the first hard drive, /dev/ttyUSB0 representing a USB serial device, or /dev/input/eventX representing input devices like keyboards or mice.

In the context of Android, accessing /dev and its contents through the adb shell command can provide insights into the device's hardware and its interactions with the Android operating system, which can be useful for troubleshooting and understanding the device's configuration.

Q03: In what partition are app APKs stored, and where is this partition mounted onto the filesystem?

Answer: In Android, app APKs (Android application packages) are typically stored in a specific partition called the "system partition." The system partition is mounted onto the filesystem at the root directory / of the Android operating system.

APK Storage: The APK files of system apps (e.g., Google apps) are stored within the system partition in the /system/app/ directory.

Mount Point: The system partition is mounted at the root directory /, which means that when you access the root directory of the Android filesystem, you are, in fact, accessing the contents of the system partition. This is why you often see system-related directories and files in the root directory when examining the Android filesystem.

```
/dev/block/dm-4=partition  
/ = mount point
```

```
C:\Users\cs2lab\AppData\Local\Android\Sdk\platform-tools  
Adb shell  
cd /system/app
```

Q04: In what filesystem (partition) do apps typically store sqlite databases? What is the /dev name of this partition?

Answer: In Android, apps typically store their SQLite databases in the user data partition, which is often referred to as the '/data/data' partition. For example, if an app has the package name "com.example.myapp," its SQLite databases would be located in:

```
/data/data/com.example.myapp/databases/dbname.db
```

where "com.example.myapp" is the package name of the app, and "dbname.db" is the name of the SQLite database file.

the /dev name of this partition is '/dev/block/dm-33'

Q05: Where is the SD-card mounted?

Answer: In Android versions, SD card is mounted at a location like '/mnt/sdcard' or '/mnt/sdcard/extsd' or '/storage/1718-1F1D'

Q06: What is the largest mounted partition in the emulator (excluding the SD-card)? What is the /dev name of this partition?

Answer: The largest mounted partition in the emulator is, excluding the SD card (if one is present), is typically the **user data partition**. This partition contains user data, app data, user settings, and user-installed apps. Actually this partition is distinct from the system partition, which contains the Android operating system and system apps.

The **/dev name** for the user data partition can vary depending on the emulator's configuration and the specific Android Virtual Device (AVD) settings you are using. However, a common **/dev name** for the user data partition in Android emulators is **/dev/block/dm-33**

Q07: Can you create files in /system? Why/why not?

Answer: By default, we cannot create files in /system partition and its read-only to prevent unauthorized changes. This partition contains very important and essential files like the core Android operating system and pre-installed system applications. If we want to create files or modify files inside this partition, we need to have root user privileges.

There are some reasons to avoid creating file or modifying system partition:

Integrity of the system: The /system partition holds the most essential and important android files like the core Android operating system and pre-installed system apps. Alteration of this partition can potentially compromise the integrity and stability of the Android system.

Updates: Android updates and security patches are often distributed by overwriting the contents of the /system partition. If users could freely modify this partition, it could interfere with the update process and potentially lead to an inconsistent or broken system.

Warranty and Support: Modifying the /system partition, including rooting the device to gain write access, often voids the warranty and can make it more challenging to receive support from the device manufacturer or service provider.

Q08: Can you create files in /data? Why/why not?

Answer: No, we cannot create files directly in the /data directory on an Android device without root privileges. The /data directory is a system directory that contains critical data and configurations for the Android operating system. It is protected for security and stability reasons, and regular apps do not have permission to write or modify files in this directory.

Here's why you cannot create files in the /data directory without root access:

Security: Allowing regular apps to write to system directories like /data would present a significant security risk. It could potentially lead to unauthorized modification of the Android operating system, causing instability or compromising the device's security.

Data Isolation: Android follows a strict security model, and each app operates within its own sandboxed environment. Apps are not permitted to access or modify each other's data or system files, including those in the /data directory.

Q09: Which filesystem do you think is most important/interesting when we do mobile forensics? Why?

Answer: When performing mobile forensics, the most important and interesting filesystems to examine typically include:

User Data Partition (often ext4 or f2fs): This partition contains the user's data, including app data, user-generated files, multimedia, and other personal information. It is a primary focus in mobile forensics because it holds valuable evidence related to the user's activities and communications.

Application Data Directory: Inside the user data partition, examining the data directories of specific apps can provide crucial insights into the user's behavior, including chat logs, images, videos, and other app-specific data.

SQLite Databases: Mobile apps frequently use SQLite databases to store structured data, making them a key source of information. Analyzing these databases can reveal chat messages, contacts, call logs, and various other app-specific data.

System Logs: Examining system logs (e.g., Android logs, iOS system logs) can provide information about system events, device activities, and potential security incidents.

External Storage (if applicable): On Android devices, external storage like the SD card can contain additional user-generated data, such as photos, videos, and downloaded files.

Cloud Data: In some cases, it may be important to investigate data stored in cloud services associated with the device, such as Google Drive or iCloud.

SMS/MMS Messages: Investigating the SMS and MMS message databases can provide insights into text and multimedia messaging history.

Call Logs: Analyzing call logs can reveal details about phone calls, including numbers dialed, call duration, and call history.

Geolocation Data: GPS data, location history, and geolocation-related files are important when examining location-based activities or tracking the movements of a mobile device.

Q10: What is the difference between physical and logical acquisition?

Answer: Physical acquisition and logical acquisition are two different methods used in digital forensics and mobile device investigations to obtain data from a device.

Scope of Data:

Physical Acquisition: Physical acquisition involves creating a bit-by-bit copy of the entire storage medium of a device, including all partitions and sectors. This means it captures all data on the device, including deleted files and system areas. It provides the most comprehensive view of the device's storage.

Logical Acquisition: Logical acquisition focuses on extracting specific data or files from a device, typically through a file system or software interfaces. It doesn't create a complete clone of the device but rather targets specific user data, apps, or files. It may not recover deleted data or system-level information.

Method and Access:

Physical Acquisition: Physical acquisition requires direct access to the device's hardware. It often involves techniques like creating a forensic image of the device's storage using specialized hardware or software tools. This method may bypass device security mechanisms and encryption, depending on the device's state.

Logical Acquisition: Logical acquisition is usually performed using software tools and interfaces provided by the operating system or manufacturer. It relies on standard communication channels (e.g., USB, Wi-Fi, or cloud services) to access data. It operates within the constraints of device security mechanisms.

Q11: What information do you lose when copying files in either of these ways?

Answer: The content of a file is frequently preserved when it is copied, but some metadata—such as permissions, creation and modification dates, and extended file attributes—may not always be preserved, particularly when the file is being copied across different file systems or storage media. The copied files might not have the same permissions as the original files when using ADB to copy them which can read, write, or execute the files may be impacted by this.

The files that have been copied might have a new owner. Usually, the files in the /data/data/com.google.android.gms directory belong to the user of the app. The person executing the ADB command may possess them after they have been copied. On Android, files may occasionally contain Access Control Lists (ACLs) that provide extra permissions. File copies might not be preserved

Q12: Is this copy-method forensically sound?

Answer: Generally speaking, the copy procedure you described—which involves pulling files from an Android device using ADB—is not regarded as forensically sound. Original data must be preserved for forensic investigations. You are effectively reading and copying data from the device when you use ADB to pull files, which entails dealing with the storage.

To guarantee that no alterations are made to the original data in a forensic setting, write protection is essential. This degree of security is not offered by ADB. Keeping an intact chain of custody is crucial to forensic investigations in order to protect the integrity of the evidence. In comparison to more specialized forensic instruments and techniques, ADB pulls might not offer the same degree of documentation and control over the evidence chain of custody.

Q13: How many packages are installed? Provide a one-line unix command line that determines this.

Answer: There are **191** packages are installed.

Command: `pm list packages |wc -l`

Output:

```
Package : com.android.se
Package : com.android.paccprocessor
Emu64x : / # pm list packages | wc -l
191
Emu64 : / #
```

Q14: Can you find the emails you sent before in the app files? In what files?

Answer: Yes, on Android system, emails sent using the Gmail app is stored in the /data/data/com.google.android.gms/ directory file. Here in lab, we go inside '/difo/1718-1F1D/com.google.android.gms' then: `grep -r " its a boy"`

Output: `grep: databases/EmailProvider.db: binary file matches`
`files/body/0/68/68.html:<meta http-equiv="Content-Type"`
`content="text/html; charset=utf-8"><div dir="auto">its a boy</div>`

Q15: What is the package name of the discord app and where does the app store its files?

Answer: The package name (also known as the Application User Model ID or AUMID) for the Discord app on Windows is: com.Discord

Command: `pm list packages`

On Windows, Discord's files are stored in the following directory: %AppData%\Discord
You can type %AppData% in the Windows Run dialog (Win + R) to open the AppData\Roaming directory, and then navigate to the Discord folder.

Q16: What files are involved storing chat messages and what are their filetypes?

Answer: Discord stores chat messages and other data in a proprietary and often encrypted format, making it challenging to directly access or manipulate this data by examining specific files. Still if we want to work with chat messages from Discord, it's recommended to use the Discord API and libraries like **discord.py**, **discord.js**, or Discord's official API.

Q17: Provide a short python script (or whatever language you prefer) that prints out the chat messages from those files, in chronological order (i.e. sorted by time). The program should take the path to the file as an argument, for example: python3 yourscripy.py thefile.bin

Answer: To access Discord chat messages using a Python script, we'll need to create a Discord bot and use the Discord API. Here's a simplified python script using the **discord.py**.

```
import discord
import asyncio

TOKEN = 'YOUR_BOT_TOKEN'

client = discord.Client()

@client.event
async def on_ready():
    print(f'Logged in as {client.user.name}')
    print('-----')

@client.event
async def on_message(message):
    if message.channel.type == discord.ChannelType.text:
        print(f'{message.author.name}: {message.content}')

async def start_bot():
    await client.start(TOKEN)

if __name__ == '__main__':
    loop = asyncio.get_event_loop()
    loop.run_until_complete(start_bot())
```

Q18: In the emulator, open google chrome and search google for something unique (e.g. "heffaklump"). Next, perform the same procedure as before (search, close, acquisition and then analysis) and see if you can find your search query in the google chrome app files. **Does hindsight find the same queries you did?**

Answer: Hindsight is a digital forensic tool that is used to analyze web browser artifacts from various web browsers, helping forensic investigators and analysts gather information related to a user's web browsing history.

After searching in Google using chrome browser a word say: Einstein Field Theory, I start 'hindsight_gui.exe' where 'http://localhost:8080/' start to listening. Then run 'localhost:8080' in hindsight and type in Profile path: '..\Users\AppData\Local\Google\Chrome\User Data\Default' and Input Type: 'Chrome'. Click on 'Run' button. Then an Output come out with 'Parsed Artifacts' and 'Plugin results' where number of URL records, Download records, Cache records, Cookie records etc. are available.

As we search "Einstein field theory" on Google in the past, Hindsight shows us a list of all of the Google search results that we clicked on, as well as the date and time of each search. Here is an example of what Hindsight shows as we search for "Einstein field theory":

Search term: Einstein Field Theory

Date: 2023-10-24 11:17:01 CST

Website: Google Search

Forensic Report (Part-2)

Case: Cat Petting and Possession of Cat Photos. Case No. [123456]

Date: 2023-11-02

Investigators (National Forensic Institute of Nocatland):

Samir Hossain Santo Bepu
Mohammad Jahid Hasan
Khondaker Refai Arafat

I. Executive Summary

This report summarizes the findings of the forensic examination of the suspect's smartphone in the case of an individual suspected of petting a real cat and possessing cat photos, as per the laws of Nocatland. The purpose of this report is to provide a comprehensive analysis of the evidence found on the smartphone and to address the questions posed by the task force.

II. Examination Methodology

The forensic examination was conducted in accordance with the standard procedures for digital forensic analysis. The smartphone was identified as Model: Samsung Galaxy S9 Duos, Android OS version: 10, Firmware version: G960FXXSHFUJ2, with the unique identifier IMEI: 35466310-213822-0, IMSI: 354663102138226, ICCID: 8946085002703845581 and Phone number: +46731421535, was received for examination. The analysis was performed using industry-standard forensic tools XAMN and XRY in Windows environment and with the help of Kali Linux.

Files preserved as Chain of Custody with evidence of integrity using hash (sha1):

- 2023-03-29_16.43_cyfo_lab1-----Samsung SM-G960f Galaxy S9 Duos TD-LTE--Google--01.xry hash is: 7e9d8dcb24ecfcb92aae3f66d1a964d8896769a
- 2023-03-29_16.43_cyfo_lab1-----Samsung SM-G960f Galaxy S9 Duos TD-LTE.xry hash is: f496d611125a58f7cc3f9df29e25b9889a60dde1
- 2023-03-29_16.43_cyfo_lab1.xrycase hash is: dce82787b840d44cad9a43331a4b913993be0b01

III. Findings and Analysis

Possession of Photos Portraying Real Cats:

- A. The smartphone contains 65 photos portraying real cats.

B. Out of the total 65 photos, 11 were taken by the phone's camera, indicating that 54 photos were either machine-learning generated or downloaded from the internet, whatsapp, telegram or gmail. There are 2 videos of petting cat and hiding of glove also recorded in suspect's mobile camera.

Machine-Learning Generated Cat Photos:

A. The smartphone contains 6 machine-learning-generated cat photos. From those only of 2 styles – detected through examine the eye-colours are different of a single cat and number of legs are 6 of another cat. We might count the photos as 2, but as of forensic purpose, we count them as 6.

Petting of a Cat:

A. Evidence of petting cats was found on the smartphone where 3 cats in multiple (atleast 4) occasions. A video clip of the suspect petting a cat wearing a sky-blue glove was taken at 27-March-2023 12:50:31 (UTC) and text messages between the suspects and accomplice. The petting photos were taken timestamps are: 26-March-2023 12:03:35 (UTC), 11:56:51 (UTC), 12:41:55 (UTC), 12:03:29 (UTC), 12:01:58 (UTC), 28-March-2023 20:20:42 (UTC), 29-March-2023 16:19:19 (UTC).

B. The specific cat involved in the act has been identified as few black spots, especially on forehead and on back on white body at KISTA NOD12 building in Security Lab from checking the geo-location of meta data. The motive behind this act is unclear. Other cats are of mostly gray colours.

C. The analysis reveals that different cats were petting on different occasions, especially twice with wearing a sky-blue glove by the suspect and twice on different cats in different time.

Suspect's Attempt to Obstruct Police:

Evidence was found to suggest that the suspect intentionally attempted to destroy or hide evidence on the smartphone. In one video it was found that the suspect hides the sky-blue glove in a cabinet of the lab. The suspect tries to fool police by deleting photos and suspicious text messages and emails related to cat petting.

Suspected Accomplices:

The forensic examination of the smartphone yield evidence of an accomplice atleast. The phone contained evidence that the suspect had been communicating with another person about suspicious of petting a cat through text-messages and emails (suspect's email

address: <fcics7581@gmail.com> and accomplice's email address:
<johannes.olegard@dsv.su.se>).

IV. Conclusion

Based on the findings of the digital forensic examination, it can be determined that the suspect had 65 photos portraying real cats, 11 of which were taken by the smartphone's camera, 2 video recordings and 6 machine-learning-generated cat photos. Evidence of petting a cat was also identified, with the act occurring on between 26-March-2023 to 29-March-2023. The different cats were petted multiple times.

The phone has evidence that the suspect had accomplice about suspicious text-messaging and emails also tries to fool police. It is essential to consider that this report is based solely on digital evidence, and further non-digital investigations may provide a more comprehensive understanding of the case.